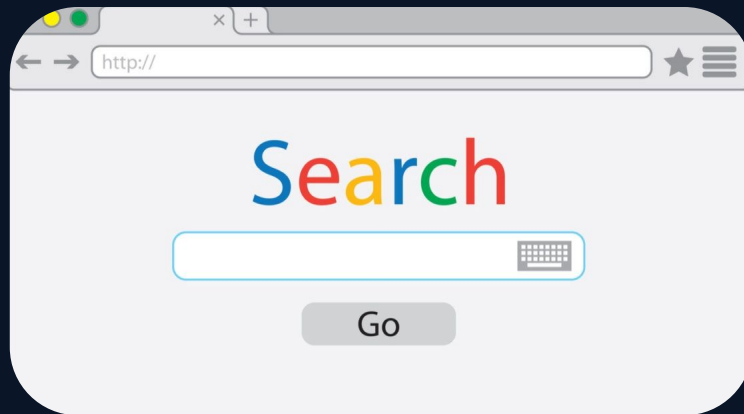


Mini Search Engine



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System Documentation

A complete web crawler and search engine built in Python, featuring TF-IDF ranking, PageRank scoring, and freshness decay.

Web

Crawl

SQLite

Index

**TF-IDF
+ PageRank**

Rank

CLI

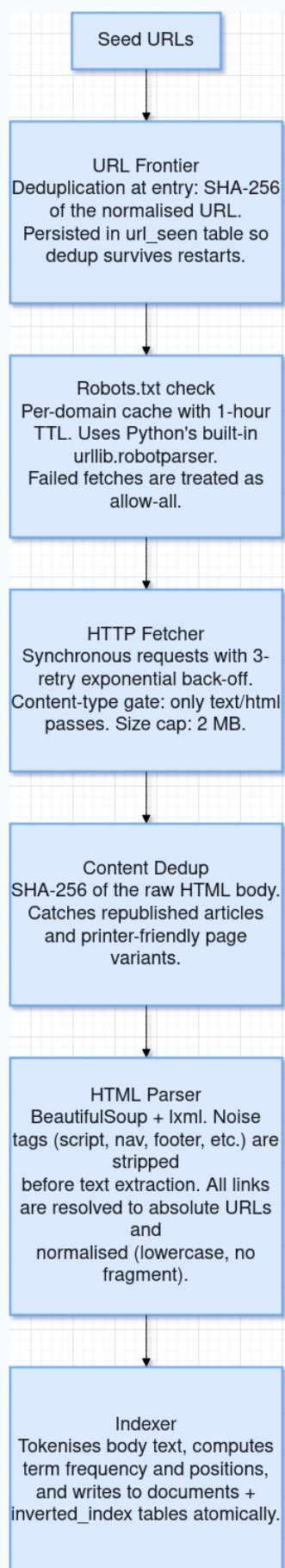
Interface

Core Components

- `crawler/crawler.py` -- Web Crawler (URL Frontier, Fetcher, Deduper, Parser, Indexer)
- `search_engine/searcher.py` -- Query Engine (Processor, Retriever, Ranker, Snippet)
- `database.sqlite` -- Shared store (documents, inverted_index, url_seen, content_seen)

Crawler - Pipeline Detail

Step-by-step processing for each crawled URL



1. URL Frontier

Deduplication at entry using SHA-256 of the normalised URL. Persisted in url_seen table so deduplication survives restarts.

2. Robots.txt Check

Per-domain cache with 1-hour TTL. Uses Python's built-in urllib.robotparser. Failed fetches treated as allow-all.

3. HTTP Fetcher

Synchronous requests with 3-retry exponential back-off. Content-type gate: only text/html passes. Size cap: 2 MB.

4. Content Dedup

SHA-256 of the raw HTML body. Catches republished articles and printer-friendly page variants.

5. HTML Parser

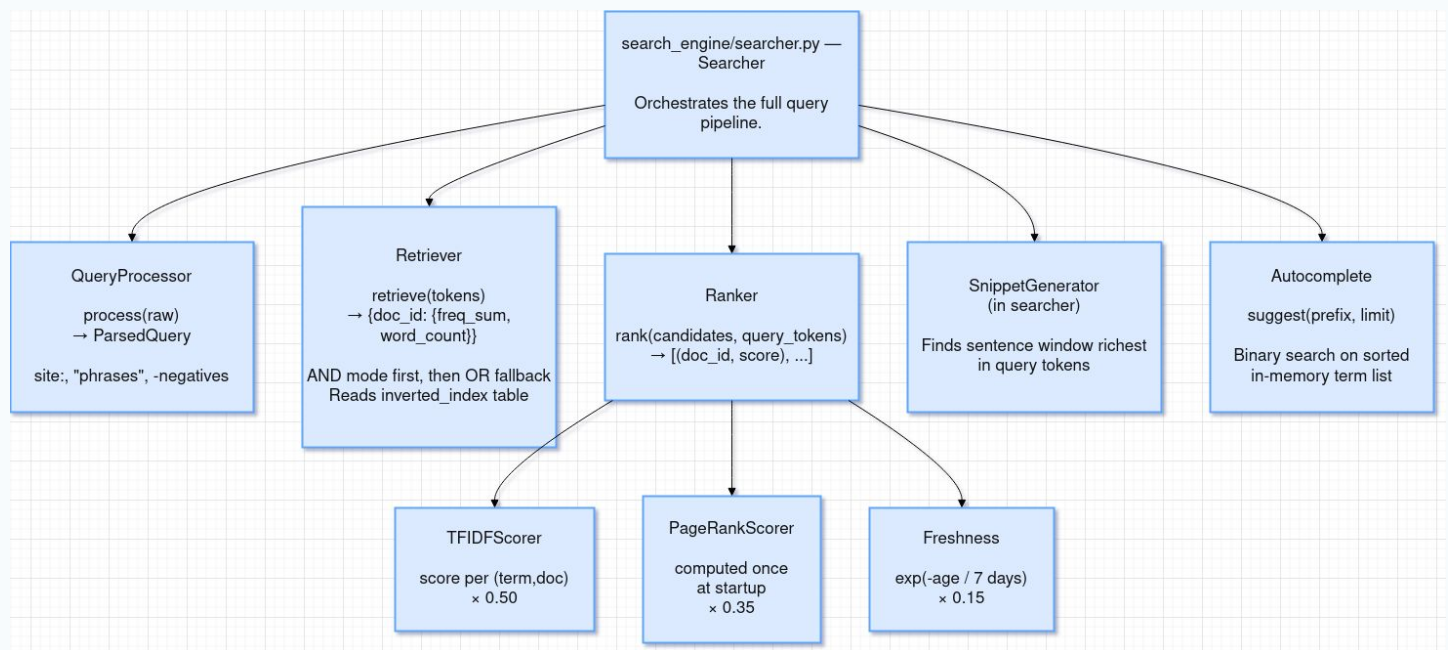
BeautifulSoup + lxml. Noise tags (script, nav, footer, etc.) are stripped before text extraction. Links resolved to absolute URLs and normalised (lowercase, no fragment).

6. Indexer

Tokenises body text, computes term frequency and positions, writes to documents + inverted_index tables atomically.

Search Engine - Component Architecture

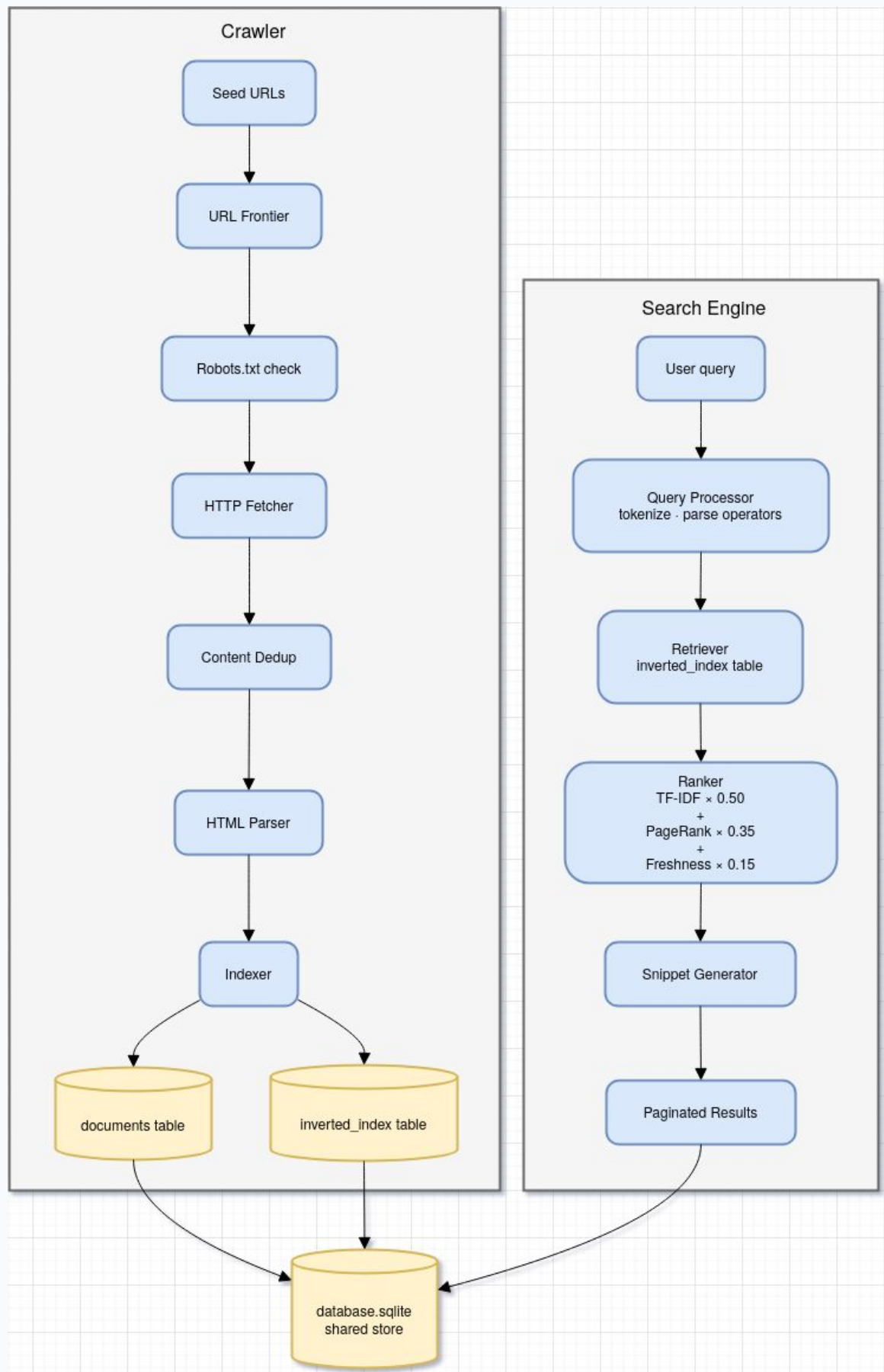
search_engine/searcher.py orchestrates the full query pipeline



QueryProcessor	Retriever	Ranker	SnippetGen + Auto
- process(raw) -> ParsedQuery - Parses site:, quoted phrases, and -negative operators Tokenises + stems using same pipeline as the Indexer	- retrieve(tokens) -> {doc_id: {freq_sum, word_count}} - AND mode first; OR fallback if AND returns nothing Reads inverted_index table directly	- rank(candidates , query_tokens) -> [(doc_id, score), ...] - Blended score: 0.50 x TF-IDF + 0.35 x PageRank + 0.15 x Freshness PageRank computed once at startup, cached in memory	- Finds sentence window with highest query-term density - Trimmed to 220 characters for display Autocomplete: binary search on sorted in-memory term list

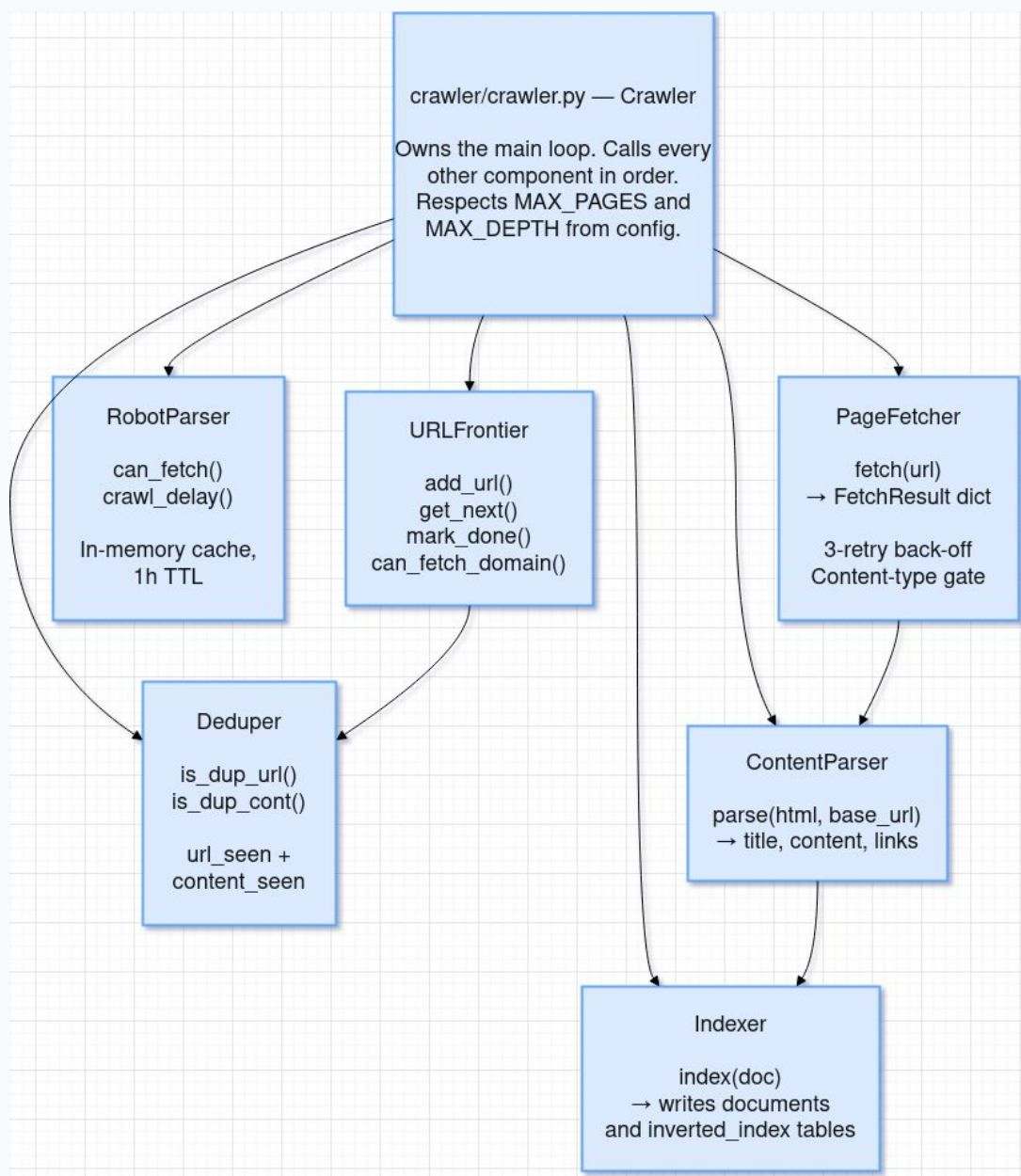
System Architecture Overview

Two main subsystems share a single SQLite database store



Crawler - Component Architecture

crawler/crawler.py orchestrates all crawl-phase components



Crawler

Main loop controller. Respects `MAX_PAGES` and `MAX_DEPTH` from config.

URLFrontier

`add_url()`, `get_next()`, `mark_done()`, `can_fetch_domain()`. Dedup at entry via SHA-256.

RobotParser

`can_fetch()` and `crawl_delay()`. In-memory cache with 1h TTL.

PageFetcher

`fetch(url)` returns `FetchResult` dict. 3-retry exponential back-off. Content-type gate: text/html only. 2 MB cap.

Deduper

`is_dup_url()` and `is_dup_cont()` using SHA-256 hashes. Uses `url_seen` + `content_seen` tables.

ContentParser

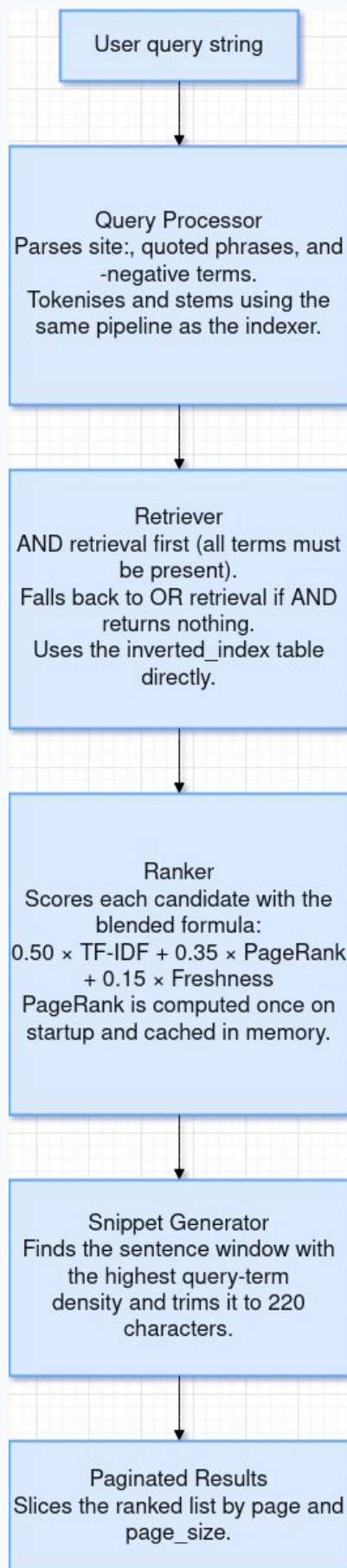
`parse(html, base_url)` returns title, content, links. Powered by BeautifulSoup + lxml. Noise tags stripped.

Indexer

`index(doc)`: tokenises body, computes TF and positions, writes to documents and inverted_index atomically.

Search Engine - Query Pipeline Detail

Step-by-step processing for each user query



1. Query Processor

Parses site:, quoted phrases, and -negative terms. Tokenises and stems using the same pipeline as the Indexer for consistency.

2. Retriever

AND retrieval first: all terms must be present in the document. Falls back to OR retrieval if AND returns nothing. Uses inverted_index table directly.

3. Ranker

Scores each candidate with: $0.50 \times \text{TF-IDF} + 0.35 \times \text{PageRank} + 0.15 \times \text{Freshness}$. PageRank is computed once on startup and cached in memory. Freshness uses $\exp(-\text{age} / 7 \text{ days})$.

4. Snippet Generator

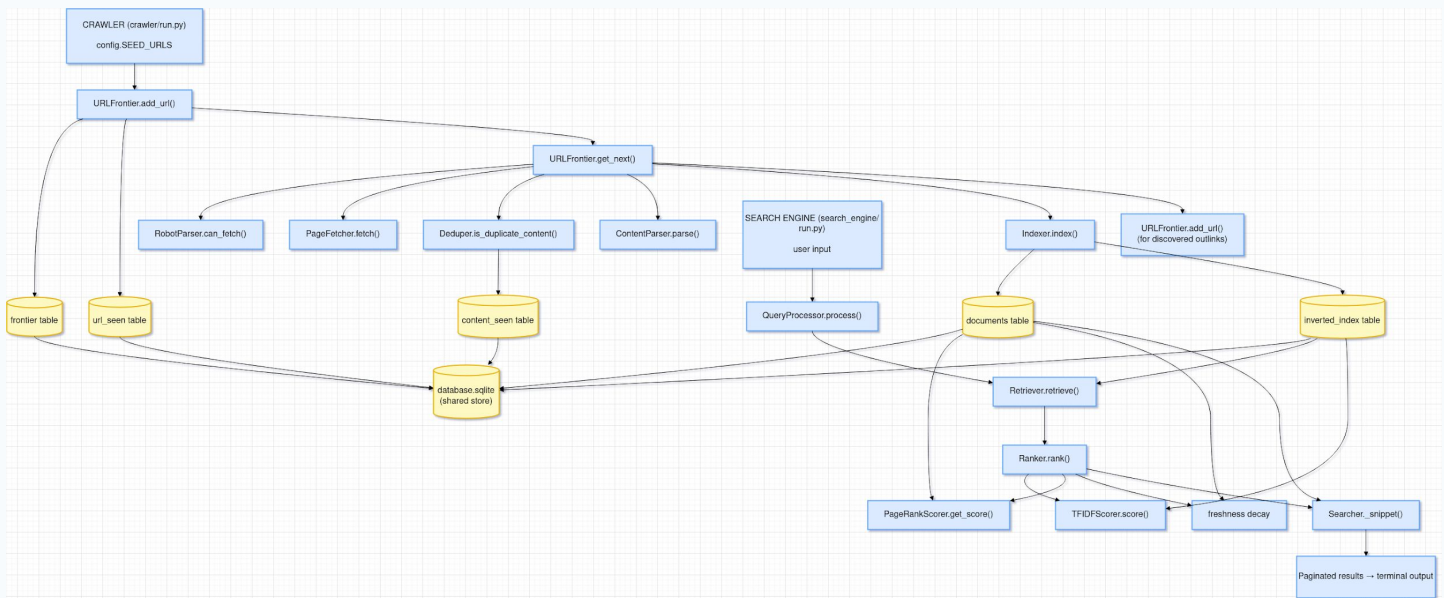
Finds the sentence window with the highest query-term density and trims it to 220 characters for compact display.

5. Paginated Results

Slices the ranked list by page and page_size. Outputs numbered results with title, URL, snippet, and relevance score to the terminal.

Full System Data Flow

End-to-end function call graph from crawl through search to output



Crawler side

CRAWLER (crawler/run.py) seeds config.SEED_URLS into URLFrontier. Each URL goes through RobotParser, PageFetcher, Deduper, ContentParser, and Indexer before any outlinks are re-queued.

Shared store

database.sqlite is the single source of truth. Both crawler and search engine read and write it. Tables: frontier, url_seen, content_seen, documents, inverted_index.

Search side

SEARCH ENGINE (search_engine/run.py) accepts user input, runs QueryProcessor.process(), then Retriever, Ranker, and Searcher.snippet() before paginating to terminal.

Scoring chain

Ranker calls PageRankScorer.get_score(), TFIDFScorer.score(), and freshness decay independently, then combines them as $0.50 + 0.35 + 0.15$ weighted sum.

Sample Search Output

Terminal results for query 'python' -- 46 results, page 1 of 5

```
Search> python

46 result(s) for 'python' [page 1/5]

1. Support the PSF with a Donation or by becoming a Supporting Member! | Python Software Foundation
https://www.python.org/psf/donations/
The official home of the Python Programming Language
score=0.4789

2. Registration Information

    - PyCon US 2026
https://us.pycon.org/2026/attend/information/
By offsetting those costs, we can offer attendees reasonable registration rates and offer a
travel grant program to those that need it. Furthermore, the revenue that PyCon US
generates is used to run the Python Software ...
score=0.4185

3. TensorFlow
https://www.tensorflow.org/
An end-to-end open source machine learning platform for everyone. Discover TensorFlow's
flexible ecosystem of tools, libraries and community resources.
score=0.4045
```

Result Fields	Scoring Breakdown	Navigation
- Rank number (1-based) - Title in cyan/green - URL in teal - Snippet (220-char window) Relevance score (0-1)	- TF-IDF x 0.50 (term relevance) - PageRank x 0.35 (link authority) - Freshness x 0.15 (recency decay) - AND retrieval; OR fallback Scores normalised to 0.0-1.0	- Results paginated (10/page default) - Shows 'N result(s) for query' - Page indicator: [page X/Y] - Type next/prev to navigate Type new query to restart